DE–3384

Sub. Code
13

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

CLASSICAL ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Prove that the sequence $\left(1+\frac{1}{n}\right)^n$ converges. (10)

   (b) Prove that the series $\sum \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \le 1$. (10)

2. (a) Test the convergence of the series $\sum \frac{1}{n(\log n)^p}$. (10)

   (b) State and prove Cauchy's root test. (10)

3. (a) Sum the series $1+\frac{3}{4}+\frac{3}{4}.\frac{5}{8}+\frac{3}{4}.\frac{5}{8}.\frac{7}{12}+\cdots$. (10)

   (b) Prove that $\lim_{x\to 0}\frac{e^{ax}-e^{bx}}{x} = a-b$. (10)

4. (a) Prove that (10)

   $$\log\frac{n+1}{n} = 2\left[\frac{1}{2n+1}+\frac{1}{3(2n+1)^3}+\frac{1}{5(2n+1)^5}+\cdots\right].$$

   (b) If $|x| < 1$ and $f(x) = x+\frac{x^3}{3}+\frac{x^5}{5}+\cdots$, prove that $f\left(\frac{2x}{1+x^2}\right) = 2f(x)$. (10)

5. (a) If α, β, γ are the roots of $x^3 + px^2 + qx + r = 0$, then find $(1+\alpha^2)(1+\beta^2)(1+\gamma^2)$. (10)

(b) Solve the equation

$8x^4 - 90x^3 + 315x^2 - 405x + 162 = 0$. Given that the roots are in Geometric progression. (10)

6. (a) Solve $6x^5 + 11x^4 - 33x^3 - 33x^2 + 11x + 6 = 0$. (10)

(b) Transform the equation $x^4 - 4x^3 - 18x^2 - 3x + 2 = 0$ into one which does not contain the third term. (10)

7. (a) Prove that $r!(2n-r)! > (n!)^2$ where $r < n$. (10)

(b) Prove that if a, b, c are three positive real numbers, then $(a+b+c)(ab+bc+ca) > 9abc$. (10)

8. (a) Find the eigen values and eigen vectors of the matrix $A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$. (10)

(b) Solve the equations :

$$x + 2y + 3z = 0$$
$$2x + y + 3z = 0$$
$$3x + 2y + z = 0$$

(10)

DE–3385

Sub. Code
14

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 =100)

1. (a) If $y = \sin(m \sin^{-1} x)$ prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$. Hence show that $y_{n+2}(0) = (n^2 - m^2)y_n(0)$. (10)

   (b) If $u = \sqrt{x^2 + y^2 + z^2}$, show that $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = 1$. (10)

2. (a) Find the radius of curvature of the curve $x^3 + 2x^2y + 3xy^2 - 4y^3 + 5x^2 - 6xy + 7y^2 - 8y = 0$. (10)

   (b) Find the envelope of the family of straight lines $\frac{x}{a} + \frac{y}{b} = 1$ where $a^2 + b^2 = k^2$. (10)

3. (a) Prove that :

   (i) $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$.

   (ii) $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$. (12)

   (b) Evaluate $\int_0^{\pi/2} \log(1 + \tan x)\,dx$. (8)

4. (a) Obtain a reduction formula for $\int x^m (\log x)^n \, dx$ and apply it to evaluate $\int_0^1 x^4 (\log x)^3 \, dx$. (10)

(b) If $I_n = \int_0^a x^n e^{-x} dx$, prove that $I_n - (n+a) I_{n-1} + a(n-1) I_{n-2} = 0$. (10)

5. (a) Solve $(2x+1)2y'' - 2(2x+1)y' - 12y = 6x$. (10)

(b) Solve $\frac{dy}{dx} - y \cot x = 2x \sin x$. (10)

6. (a) Solve $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} - 5y = \sin(\log x)$. (12)

(b) Solve $(D^2 + 3D + 2)\, y = x^2$. (8)

7. (a) Find $L^{-1}\left[\frac{2(s+1)}{(s^2+2s+5)^2}\right]$. (12)

(b) Find $L\left[\frac{1-\cos x}{x}\right]$. (8)

8. (a) Solve $(p^2 + q^2) y = qz$. (10)

(b) Solve $z = px + qy + c\sqrt{1 + p^2 + q^2}$. (10)

DE–3386

Sub. Code
15

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

ANALYTICAL GEOMETRY AND VECTOR CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

(5 × 20 = 100)

1. (a) Find the locus of P such that $PA^2 + PB^2 = K^2$ where A is $(3, 4, 5)$, and B is $(-1, 3, -7)$ and K is constant. (10)

   (b) Show that two of the straight lines $ax^3 + bx^2y + cxy^2 + dy^3 = 0$ will be perpendiculars to each other if $a^2 + d^2 = -(bd + ac)$. (10)

2. (a) Find the Cartesian equation equivalent of the polar equation $r\cos\theta = -4$. (10)

   (b) Find the circles which cut orthogonally each of the following circles : (10)

   $x^2 + y^2 + 2x + 4y + 1 = 0$

   $x^2 + y^2 - 4x + 3 = 0$

   $x^2 + y^2 + 6y + 5 = 0$

3. (a) Find the angle between the planes $2x - y + z = 6$ and $x + y + 2z = 7$. (10)

   (b) Find the root of the perpendicular from the origin on the line $3x - y - z - 4 = 0 = 4x - 3y - 2z + 2$. (10)

4. (a) Find the image of the point $(2, 3, 4)$ under the reflection in the plane $x-2y+5z=0$. (10)

(b) Find the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$. Are they coplanar? (10)

5. (a) Find the shortest distance between the lines $\frac{x-5}{3}=\frac{y-6}{-4}=\frac{z-9}{1}$ and $2x-2y+z-3=0=2x-y+2z-9$. (10)

(b) Find the equation of the plane determined by the parallel lines $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x-3}{1}=\frac{y-2}{2}=\frac{z-1}{3}$. (10)

6. (a) Prove that the equation of the tangent plane to the sphere $x^2+y^2+z^2+2ux+2vy+2wz+d=0$ at $P(x_1, y_1, z_1)$ is

$$xx_1+yy_1+zz_1+u(x+x_1)+v(y+y_1)+w(z+z_1)+d=0.$$

(10)

(b) Find the equation of the sphere passing through the circle $x^2+y^2+z^2-4=0, 2x+4y+6z-1=0$ and having its centre on the plane $x+y+z=6$. (10)

7. (a) If r is the position vector of any point $P(x,y,z)$, prove that $grad\, r^n = n.r^{n-2}\bar{r}$. (10)

(b) Evaluate $\int_0^1 \int_x^1 (x^2 + y^2)\, dy\, dx$. (10)

8. Verify Gauss divergence theorem for $\bar{f} = (x^2 - yz)\bar{i} + (y^2 - zx)\bar{j} + (z^2 - xy)\,\bar{k}$ taken over the rectangular parallelopiped, $0 \le x \le a, 0 \le y \le b, 0 \le z \le c$. (20)

DE–3387

Sub. Code
23

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

MECHANICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) State and prove triangle law of forces and also discuss the converse of the triangle of forces. (10)

   (b) *ABC* is a given triangle. Forces *P, Q, R* acting along the lines *OA, OB, OC* are in equilibrium. Prove the following : (10)

   (i) $P : Q : R = a^2(b^2 + c^2 - a^2) : b^2(c^2 + a^2 - b^2) : c^2(a^2 + b^2 - c^2)$

   if *O* is the circumcentre of the triangle.

   (ii) $P : Q : R = \cos\frac{A}{2} : \cos\frac{B}{2} : \cos\frac{C}{2}$ if *O* is the incentre of the triangle.

2. (a) *ABCDEF* is a regular hexagon and at *A*, act forces represented by $\overline{AB}$, $2\overline{AC}$, $3\overline{AD}$, $4\overline{AE}$ and $5\overline{AF}$. Show that the magnitude of the resultant is $AB \cdot \sqrt{351}$ and it makes an angle $\tan^{-1}(7/\sqrt{3})$ with *AB*. (10)

(b) E is the middle point of the side CD of a square $ABCD$. Forces $16, 20, 4\sqrt{5}, 12\sqrt{2}$ kg.wt. act along AB, AD, AE, CA in the directions indicated by the other of the letters. Show that they are in equilibrium. (10)

3. (a) State and prove Varignon's theorem of moments.(10)

(b) A beam of weight W hinged at one end is supported at the other end by a string so that the beam and the string are in a vertical plane and make the same angle θ with the horizon. Show that the reaction at the hinge is $\frac{W}{4}\sqrt{8+\operatorname{cosec}^2\theta}$. (10)

4. (a) What is meant by friction? Explain. Also write down the laws of friction. (10)

(b) A uniform ladder is in equilibrium with one end resting on the ground and the other against a vertical wall; if the ground and wall be both rough, the coefficients of friction being μ and μ' respectively and if the ladder be on the point of slipping at both ends, show that θ, the inclination of the ladder to the horizon is given by $\tan\theta = \frac{(1-\mu\mu')}{2\mu}$. (10)

5. (a) Show that the greatest height which a particle with initial velocity v can reach on a vertical wall at a distance 'a' from the point of projection is $\frac{v^2}{2g} - \frac{ga^2}{2v^2}$.

Also prove that the greatest height above the point of projection attained by the particle in its flight is $\frac{v^6}{2g(v^4+g^2a^2)}$. (10)

(b) A particle is projected at an angle α with a velocity u and it strikes up an inclined plane of inclination β at right angles to the plane. Prove that :

(i) $\cot\beta = 2\tan(\alpha - \beta)$;

(ii) $\cot\beta = \tan\alpha - 2\tan\beta$. (10)

6. (a) A smooth circular table is surrounded by a smooth rim whose interior surface is vertical. Show that a ball projected along the table from a point A on the rim in a direction making an angle θ with the radius through A will return to the point of projection after two impacts if $\tan\alpha = \dfrac{e^{(3/2)}}{\sqrt{1+e+e^2}}$.

Also prove that, when the ball returns to the point of projection, its velocity is to its original velocity as $e^{(3/2)} : 1$. (10)

(b) Obtain the loss of kinetic energy due to direct impact of two smooth spheres. (10)

7. (a) Show that composition of two simple harmonic motions of the same period and in the same straight line. (10)

(b) A pendulum which beats seconds at the surface of the earth loses 10 seconds in 24 hours when taken to the summit of a hill. Find the height of the hill, taking the radius of the earth to be 6400 km. (10)

8. (a) Derive the differential equation of central orbit. (10)

(b) A particle moves in an ellipse under a force which is always directed towards its focus. Find the law of force, the velocity at any point of the path and its periodic time. (10)

DE–3388

Sub. Code
24

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Let (M, d) be a metric space. Define $d_1(x, y) = \dfrac{d(x, y)}{1 + d(x, y)}$. Prove that d_1 is a metric on M. (10)

   (b) Define open ball with an example. Also prove that in any metric space the intersection of a finite number of open sets is open. (10)

2. (a) Let (M, d) be a metric space and let $A, B \subseteq M$. Prove the following :

   (i) A is open if and only if $A = \text{Int } A$.

   (ii) Int A = Union of all open sets contained in A.

   (iii) Int A is an open subset of A and if B is any other open set contained in A then $B \subseteq \text{Int } A$

   (iv) $A \subseteq B \Rightarrow \text{Int } A \subseteq \text{Int } B$. (10)

   (b) Let M be a metric space and $A \subseteq M$. Prove that $\overline{A} = A \cup D(A)$. (10)

3. (a) Prove that M is connected if and only if every continuous function $f : M \to \{0, 1\}$ is not onto. (10)

(b) With the usual notations, prove that l_2 is complete. (10)

4. (a) State and prove Cantor's intersection theorem. (10)

(b) Prove that any complete metric space is of second category. (10)

5. (a) Let M be a metric space and let $A \subseteq M$. Prove that A is compact if and only if given a family of open sets $\{G_\alpha\}$ in M such that $\bigcup G_\alpha \supseteq A$ there exists a subfamily $G_{\alpha_1}, G_{\alpha_2}, \ldots G_{\alpha_n}$ such that $\bigcup_{i=1}^{n} G_{\alpha_i} \supseteq A$. (10)

(b) Show that continuous image of a compact metric space is compact. (10)

6. (a) Let (M_1, d_1) and (M_2, d_2) be two metric spaces. Prove that $f : M_1 \to M_2$ is continuous if and only if $f^{-1}(G)$ is open in M_1 whenever G is open in M_2. (10)

(b) Define an uniformly continuous function. Also prove that the function $f : R \to R$ defined by $f(x) = \sin x$ is uniformly continuous on R. (10)

7. (a) State and prove Dini's theorem. (10)

(b) State the Weierstrass M-test. Also prove that the series $\sum_{n=1}^{\infty} \frac{nx^2}{n^2 + x^2}$ is uniformly convergent on $[0, k]$ where $k > 0$. (10)

8. (a) (i) Define a contraction mapping.

(ii) $T : l_2 \to l_2$ defined by $T(x) = \frac{1}{2} x_n$. Prove that T is a contraction mapping where $x = (x_n)$.

(iii) Let $T : M \to M$ be a contraction mapping. Prove that T is continuous on M. (10)

(b) State and prove the Piccard's theorem. (10)

DE-3389

Sub. Code
25

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

PROBABILITY AND STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) If A and B are any two events then prove that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. (10)

   (b) An integer is chosen at random from the first two hundred integers, what is the probability that the integer chosen is divisible by 6 or 8. (10)

2. (a) Calculate the mean and standard deviation for the following : (10)

x :	20-30	30-40	40-50	50-60	60-70	70-80	80-90
f :	3	61	132	153	140	51	2

   (b) Obtain Karl Pearson's measures of Skewness for (10)

x :	5-10	10-15	15-20	20-25	25-30	30-35	35-40
f :	6	8	17	21	30	35	40

3. (a) Let the random variables X and Y have the joint p.d.f. $f(x,y) = \begin{cases} x+y, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$.

   Compute the correlation coefficient of X and Y. (10)

   (b) If the correlation coefficient of X and Y exists, show that $-1 \le p \le 1$. (10)

4. Ten competitors in a musical test were ranked by the three judges x, y, z in the following data :

Rank by x :	1	6	5	10	3	2	4	9	7	8
Rank by y :	3	5	8	4	7	10	2	1	6	9
Rank by z :	6	4	9	8	1	2	3	10	5	7

Using rank correlation method, discuss which pair of judges has the nearest approach to common likings in music. (20)

5. (a) X is a random variable following binomial distribution with mean 2.4 and variance 1.44. Find $P(x \geq 5)$ and $P(1 < X \leq 4)$. (10)

(b) If the probability of success is $\frac{1}{100}$, how many trials are necessary in order that probability of atleast one success is greater than $\frac{1}{2}$. (10)

6. (a) In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. (10)

(b) A random variable X has probability function $P(x) = \frac{1}{2^x}, x = 1,2.....$ Find the moment generating function, mean and variance. (10)

7. Derive Fisher's F distribution. (20)

8. Analyse the variance in the following Latin squares : (20)

B20	C17	D25	A34
A23	D21	C15	B24
D24	A26	B21	C19
C26	B23	A27	D22

DE–3390

Sub. Code
31

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) For any three set A, B and C, prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. (10)

   (b) For any three sets A, B and C, prove that $A \Delta (B \Delta C) = (A \Delta B) \Delta C$. (10)

2. (a) Let $F : A \to B, g : B \to C$ be bijections. Prove that $g \circ f : A \to C$ is also a bijection. (10)

   (b) Show that $f : \mathbb{R} \to (0,1)$ defined by $f(x) = \frac{1}{2}\left[1 + \frac{x}{1+|x|}\right]$ is a bijection. (10)

3. (a) Give an example of a subgroup which is not a group. (10)

   (b) Prove that any permutation can be expressed as a product of disjoint cycles. (10)

4. (a) Define cyclic group. Give an example. Also show that any cyclic group is abelian. (10)

   (b) State and prove Lagrange's theorem on group. (10)

5. (a) Show that the set R of all matrices of the form $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ where $a, b \in R$ is a ring under matrix addition and matrix multiplication. (10)

(b) Prove that Z_n is a field iff n is prime. (10)

6. (a) Prove that the characteristic of an integral domain D is either 0 or a prime number. (10)

(b) Define Euclidean ring.

Prove that any Euclidean ring R has an identify element. (10)

7. (a) Define vector space.

Prove that $\mathbb{R}$ is not a vector space over $\mathbb{C}$. (10)

(b) Let V and W be vector spaces over a field F.let $T : V \to W$ be an isomorphism. Prove that T maps a basis of V onto a basis of W. (10)

8. Prove that every finite dimensional inner product space has an orthonormal basis. (20)

DE–3391

Sub. Code
32

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Solve the following LPP graphically : (10)

Maximize $z = 100x_1 + 40x_2$

Subject to :

$5x_1 + 2x_2 \leq 1000$
$3x_1 + 2x_2 \leq 900$
$x_1 + 2x_2 \leq 500$
and $x_1, x_2 \geq 0$

(b) Rewrite in standard form the following LPP : (10)

Maximize $z = 2x_1 - x_2 + x_3$

Subject to the constraints

$x_1 + 3x_2 - x_3 \leq 20;$
$2x_1 - x_2 + 3x_3 \geq 12;$
$x_1 - 4x_2 - 4x_3 \geq 2;$
$x_1, x_2 \geq 0, x_3$ is unrestricted

2. (a) Use simplex method to solve the following LPP : (10)

Maximize $z = 10x_1 + x_2 + 2x_3$;

Subject to :

$x_1 + x_2 - 2x_3 \leq 10;$
$4x_1 + x_2 + x_3 \leq 20;$
$x_1, x_2 \geq 0$

(b) Write the dual simplex algorithm. (10)

3. (a) Solve the following LPP using Big-M method : (10)

Minimize : $z = 60x_1 + 80x_2$

Subject to :

$20x_1 + 30x_2 \geq 900$
$40x_1 + 30x_2 \geq 1200$.
$x_1, x_2 \geq 0$

(b) Write the simplex algorithm for two phase method. (10)

4. (a) Find the integer solution to the LPP : (10)

Maximize : $z = 2x_1 + 2x_2$

Subject to :

$5x_1 + 3x_2 \leq 8$
$x_1 + 2x_2 \leq 4$
$x_1, x_2 \geq 0$ and are integers

(b) Explain the cutting plane algorithm. (10)

5. (a) Solve the following transportation problem : (10)

	1	2	3	4	Supply
A	270	230	310	690	100
B	100	450	400	320	80
C	300	540	350	570	80
Demand	60	120	50	40	

(b) Using North-West corner rule, find an initial basic feasible solution for the following T.P. (10)

	W_1	W_2	W_3	a_1
F_1	2	7	4	5
F_2	3	3	1	8
F_3	5	4	7	7
F_4	1	6	2	14
b_1	2	9	18	34
				29

6. (a) Using VAM, find a basic feasible solution to the following T.P. (10)

	1	2	3	4	a_i
I	21	16	25	13	11
II	17	18	14	23	13
III	32	27	18	41	19
b_j	6	10	12	15	43

(b) Solve the assignment problem given below : (10)

	M_1	M_2	M_3	M_4
J_1	5	7	11	6
J_2	8	5	9	6
J_3	4	7	10	7
J_4	10	4	8	3

7. (a) Solve the following game using LPP method : (10)

$$\begin{array}{cc} & \text{Player B} \\ \text{Player A} & \begin{pmatrix} 1 & -1 & 3 \\ 3 & 5 & -3 \\ 6 & 2 & -2 \end{pmatrix} \end{array}$$

(b) Verify whether the following two person zero-sum games are strictly determinable and fair. Give the optimum strategies for each player in the case of strictly determinable game (if any) (10)

$$\begin{array}{ccc} & & \text{Player B} \\ \text{Player A :} & \begin{array}{c} A_1 \\ A_2 \end{array} & \begin{pmatrix} 0 & 2 \\ -1 & 4 \end{pmatrix} \end{array}$$

8. (a) Solve graphically the game whose pay-off matrix is

$$\begin{bmatrix} 2 & 3 & 11 \\ 7 & 5 & 2 \end{bmatrix}.$$

(10)

(b) Explain the PERT algorithm. (10)

DE–3392

Sub. Code
33

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016

NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Find the smallest positive root of the equation

$x^3 - 2x + 0.5 = 0$ by Newton's Raphson method. (10)

(b) Using iteration method find a root of $x^3 - x - 4 = 0$.(10)

2. (a) Solve the system of equations :

$$\begin{aligned} 28x + 4y - z &= 32 \\ x + 3y + 10z &= 24 \\ 2x + 17y + 4z &= 35 \end{aligned}$$

, using Gauss elimination method. (10)

(b) Using Gauss-Jordan method solve the following equations :

$$\begin{aligned} 10x + y + z &= 12 \\ 2x + 10y + z &= 13 \\ x + y + 5z &= 7. \end{aligned}$$

(10)

3. (a) Prepare a forward difference table from the following table and express y as a function of x. (10)

x :	0	1	2	3	4
y :	7	10	13	22	43

(b) By using Newton's backward difference formula, find the value of y when $x = 10$ from the following :

x :	5	6	9	11
y :	12	13	14	16

(10)

4. (a) Use Bessel's formula to find $f(25)$ given that $f(20) = 24, f(24) = 32, f(28) = 35, f(32) = 40$. (10)

(b) Using Stirling's formula, evaluate $f(1.22)$ given that,

x :	0	1.0	1.1	1.2	1.3	1.4
$f(x)$:	0	0.841	0.891	0.932	0.963	0.985

(10)

5. (a) Evaluate $\int_0^1 \frac{dx}{1+x^2}$, using Trapezoidal rule with $h = 0.2$. Hence determine the value of π. (10)

(b) Find the first and second derivatives of the function tabulated below at the point $x = 1.5$ (10)

x :	1.5	2.0	2.5	3.0	3.5	4.0
$f(x)$:	3.375	7.0	13.625	24.0	38.875	59.0

6. (a) Evaluate $\int_0^2 \frac{dx}{x^3+x+1}$ to 3 decimals dividing the range of integration into 8 equal parts using Simpson's rule. (10)

(b) Derive Weddle's rules for integration. (10)

7. Find the value of $y(1.1)$ and $y(1.2)$ from $\frac{d^2y}{dx^2} + y^2\frac{dy}{dx} = x^3$; $y(1) = 1, y'(1) = 1$ by using Taylor series method. (20)

8. By applying the fourth order Runge-Kutta method, find $y(0.2)$ from $y' = y - x, y(0) = 2$ taking $h = 0.1$. (20)

DE-3393

Sub. Code
34

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) For any two complex numbers z_1 and z_2, prove that $|z_1 + z_2| \le |z_1| + |z_2|$. (10)

   (b) State and prove Cauchy-Riemann equations in polar co-ordinates. (10)

2. (a) If $f(z)$ is analytic, then prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 4|f'(z)|^2$. (10)

   (b) Prove that every power series represents an analytic function at all points within the circle of convergence and its derivative can be obtained by termwise differentiation of the given power series. (10)

3. (a) Define conformal mapping and determine the angle of rotation and scale factor at the point $z = 1 + i$ under the mapping $w = z^2$. (10)

   (b) Find the bilinear transformation which maps the points $z_1 = 0$, $z_2 = 1$, $z_3 = \infty$ on to $w_1 = i$, $w_2 = -1$, $w_3 = -i$ respectively. (10)

4. (a) Find the image of the strip $2 < x < 3$ under $w = \frac{1}{z}$. (10)

(b) Discuss the transformation $w = e^z$. (10)

5. (a) Let f be analytic at all points inside and on a simple closed curve C. Then prove that $\int_C f(z)\,dz = 0$. (10)

(b) Prove that $\int_C \frac{zd^2}{z^2 - 1} = 2\pi i$ where C is the positively oriented circle $|z| = 2$. (10)

6. (a) State and prove fundamental theorem of algebra. (10)

(b) Expand ze^{2z} in a Taylor's series about $z = -1$ and determine the region of convergence. (10)

7. (a) Expand $f(z) = \frac{-1}{(z-1)(z-2)}$ in a Laurent's series valid for $1 < |z| < 2$. (10)

(b) Define isolated singularity. Prove that an isolated singularity a of $f(z)$ is a pole if and only if $\lim_{z \to a} f(z) = \infty$. (10)

8. (a) State and prove Rouche's theorem. (10)

(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{5 + 4\sin\theta}$. (10)

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DE–3394

Sub. Code
35

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries equal marks.

$(5 \times 20 = 100)$

1. (a) (i) Define the following terms with an example for each :

Disjunction and Biconditional statement.

(ii) Draw the parsing tree for the formula

$((P \to (\neg q)) \to (p \wedge q))$. (5 + 5=10)

(b) Show that $Q \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$ is a tautology. (10)

2. (a) Obtain a disjunctive normal form of $\neg(P \vee Q) \leftrightarrow (P \wedge Q)$. (10)

(b) Find the principal disjunctive normal form of $P \to ((P \to Q) \wedge \neg(\neg Q \vee \neg P))$. (10)

3. (a) Show that $P \to S$ can be derived from the premises $\neg P \vee Q, \neg Q \vee R, R \to S$. (10)

(b) Narrate the following terms with an example for each. Open statements and quantifiers. (10)

4. (a) Show that $(\forall x)(P(x)) \vee (\forall x)(Q(x)) \rightarrow (\forall x)(P(x) \vee Q(x))$ is logically valid. (10)

(b) Show that $(\forall x)(P(x) \vee Q(x)) \Rightarrow (\forall x)P(x) \vee (\exists x)Q(x)$. (10)

5. (a) Define the following terms. Give an example for each.

(i) Complete graph

(ii) Bipartite graph

(iii) Isomorphic graph

(iv) Adjacency matrix

(v) Digraph. (10)

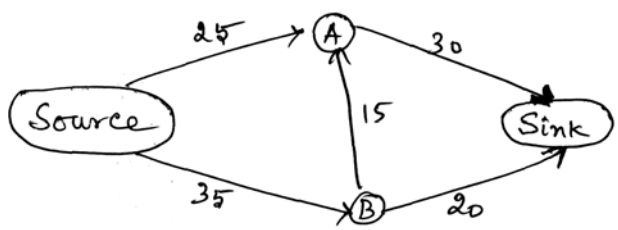
(b) Prove that a tree with n vertices has $n-1$ edges. (10)

6. (a) Using matrix tree theorem find $\tau(k_n)$, where k_n is the complete graph with n vertices. (10)

(b) Let T be a full binary tree with n vertices. Prove that the number of leaves in T is $\frac{(n+1)}{2}$. (10)

7. (a) State and prove the Dirac theorem. (10)

(b) Find a maximum flow for the following network. (10)



8. (a) (i) Show that the graph $k_{3,3}$ is not planar.

(ii) If G is a simple planar graph with atleast three vertices $e(G) \leq 3n(G) - 6$. (5 + 5 = 10)

(b) State and prove Euler's formula. (10)